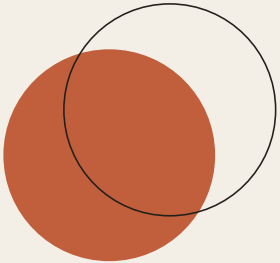
November 9, 2026

# Research with AI Agents

*What Works, What It Costs, and What an Institute Needs*

Lukas Theissinger



# Outline

I. OpenAI and Agent Usage

II. Owning Intelligence and Infrastructure at Lamarr

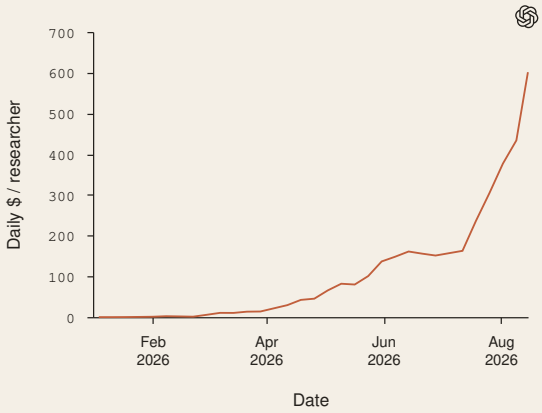


OUTLINE

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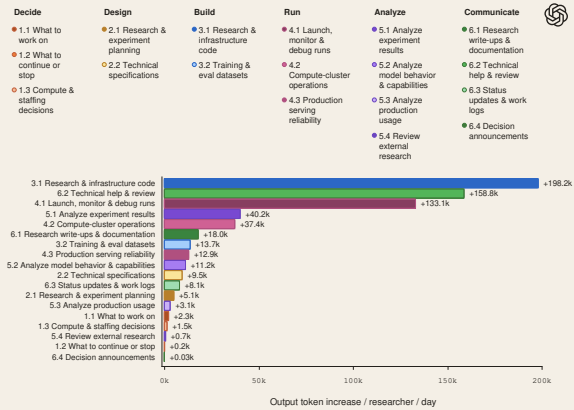
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# FOMO Headline: OpenAI Agent Usage



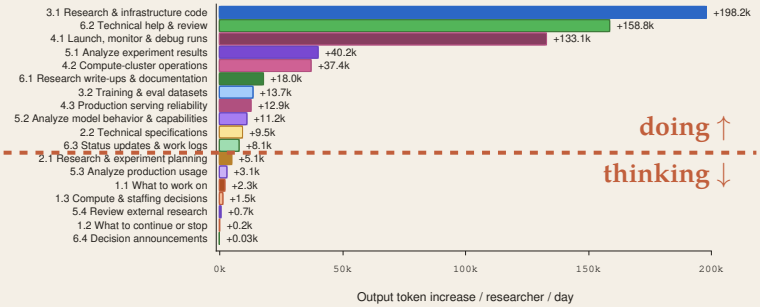

Median researcher, daily $ spent on internal coding agents. Source: OpenAI, "Research acceleration" (self-reported)

# What do agents do at OpenAI?




Increase in output tokens per researcher per day, by research activity. Source: OpenAI, “Research acceleration” (self-reported)

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| Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure | Agent use is a *skill*: not everything can or should be automated (degree of automation) |

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# Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every two weeks:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
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4. **Ask your agent to try it:** `github.com/Newida/AutomatingExercises` **Avoid:** Don't do anything BY HAND on this project. If you want to change it, ASK YOUR AGENT to change it

# II

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Collect receipts, fill in the university form, look up per-diem rules, fix what admin sends back: say **3 trips** × **3 h per person per year**. An agent that reads the receipts and files the form correctly needs ∼**200 h** (rules, forms, access to the admin system)

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| Lamarr, ∼300 researchers | $300 \times 9\,\text{h} = 2\,700\,\text{h/year}$ | 200 h | **pays >13×** |

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So let's talk about costs

# Decision 1: Subscription or API?

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- \+ flat price, no surprise bills
- \+ very cheap for heavy interactive use
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One month of my usage, priced at API list prices

| | subscription | same usage via API | factor |
|---|---|---|---|
| Claude | 20 $ (Pro) | ∼680 $ | ∼34× |
| Codex | 20 $ (Plus) | ∼145 $ | ∼7× |

# Decision 2: Self-Hosting?

In the news

> “Revolut turns to self-hosted AI to cut coding costs tenfold over Claude”

Computing (UK), 2026

# Decision 2: Self-Hosting? A Field Report: Pros and Cons

| Hardware | Model | Quality | Speed |
|---|---|---|---|
| **4 × H100** | **GLM 5.3 Flash** | **Sonnet high** | **50–100 tokens/s** |
| repurposed | mainly | how it feels | per user |

**What I would use it for**

- Pipelines, e.g. **AutoExercises**, finding related work
- High token, lower intelligence: Code implementation, Organizing experiments
- **Private Data**
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**What I would not use it for**

- High-intelligence, low-token work: Math proofs, idea finding
- Remember: OpenAI usage diagram

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**Rough, quickly made estimates**

Only to get a feeling for the headline. Energy and the Sonnet comparison are true self-hosting values; pros, cons and full costs are not thought through. Repurposing hardware also takes skill and neat tricks.

| **GLM-5.3-Flash (main model)** | **per month** |
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| energy (full load) | ∼330–670 € |
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**Payback:** 4 × H100 at ∼150 k€ pays for itself in ∼10–22 months

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